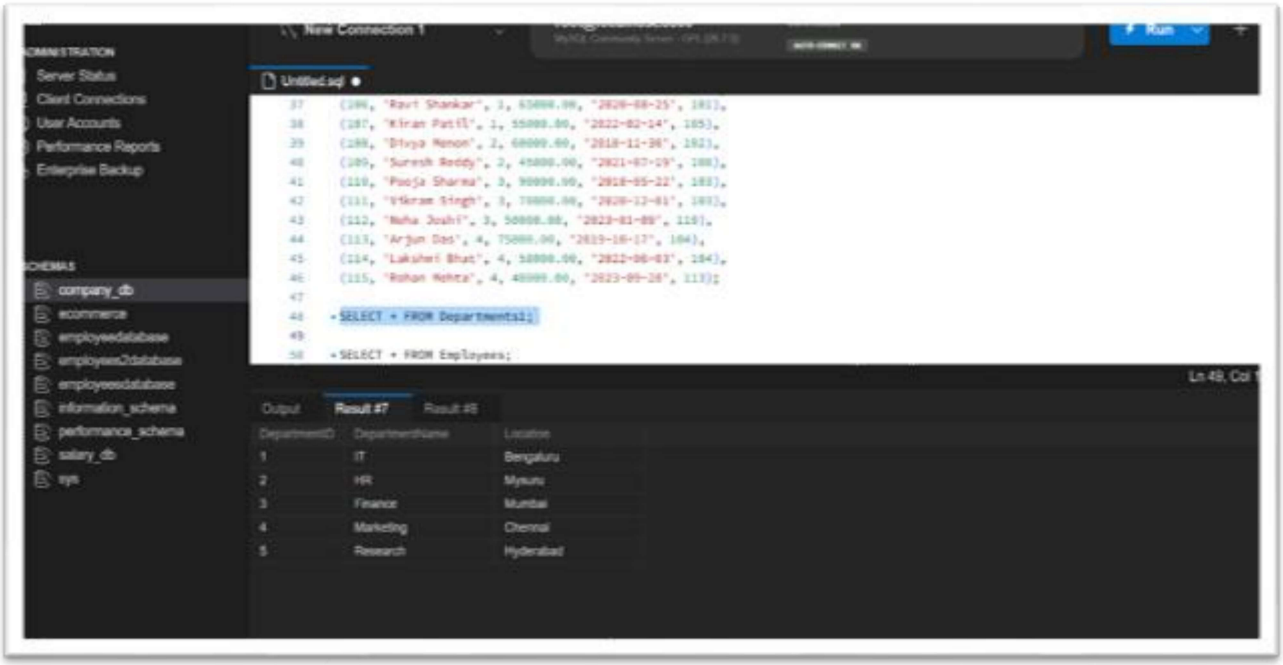
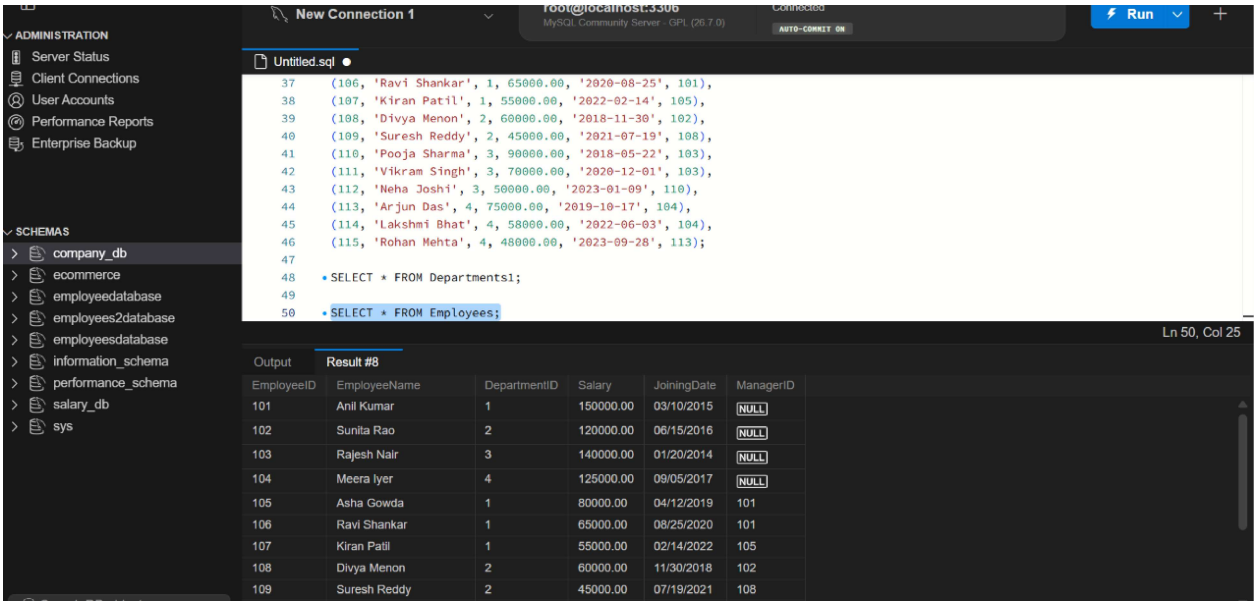
Name:Talla Tejaswanth,

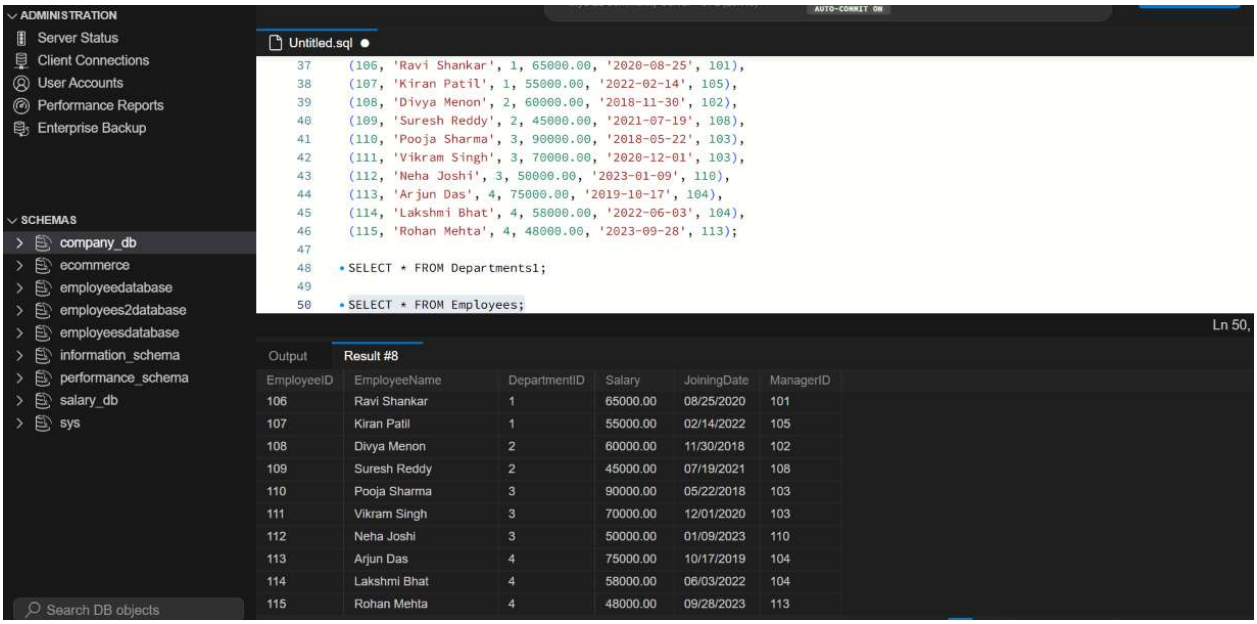
HCL ID:HCLTFP2288095

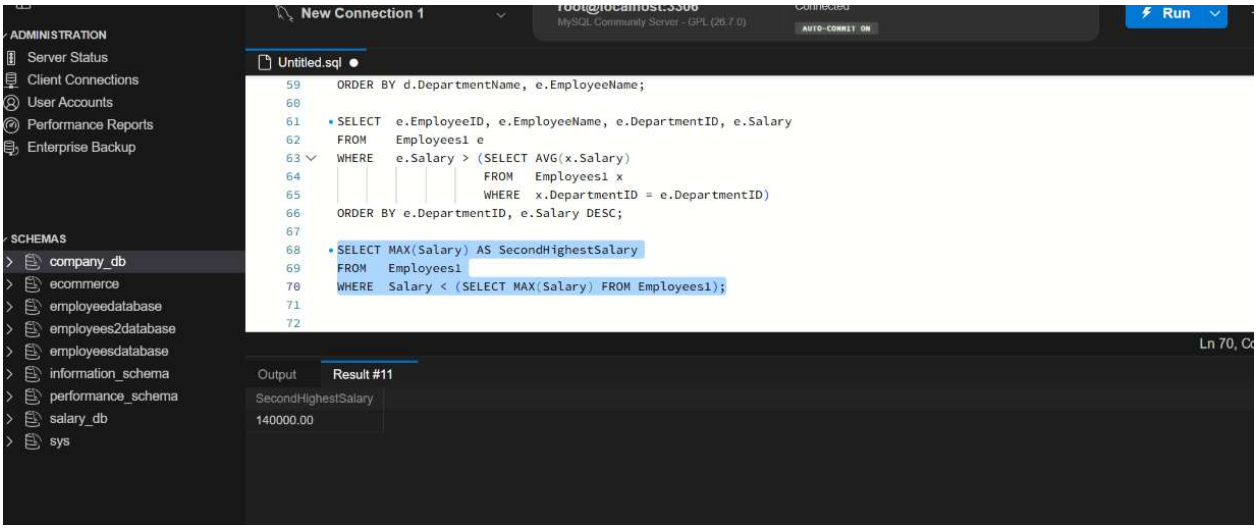
USN:23BTRCT133

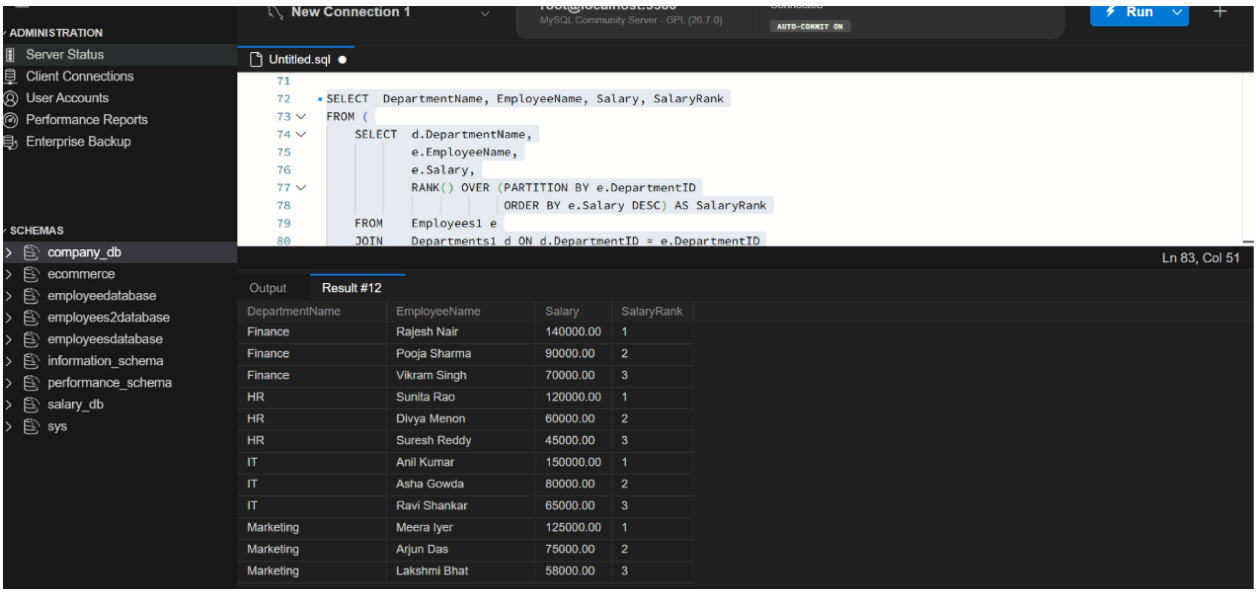
DEPT:CSE*

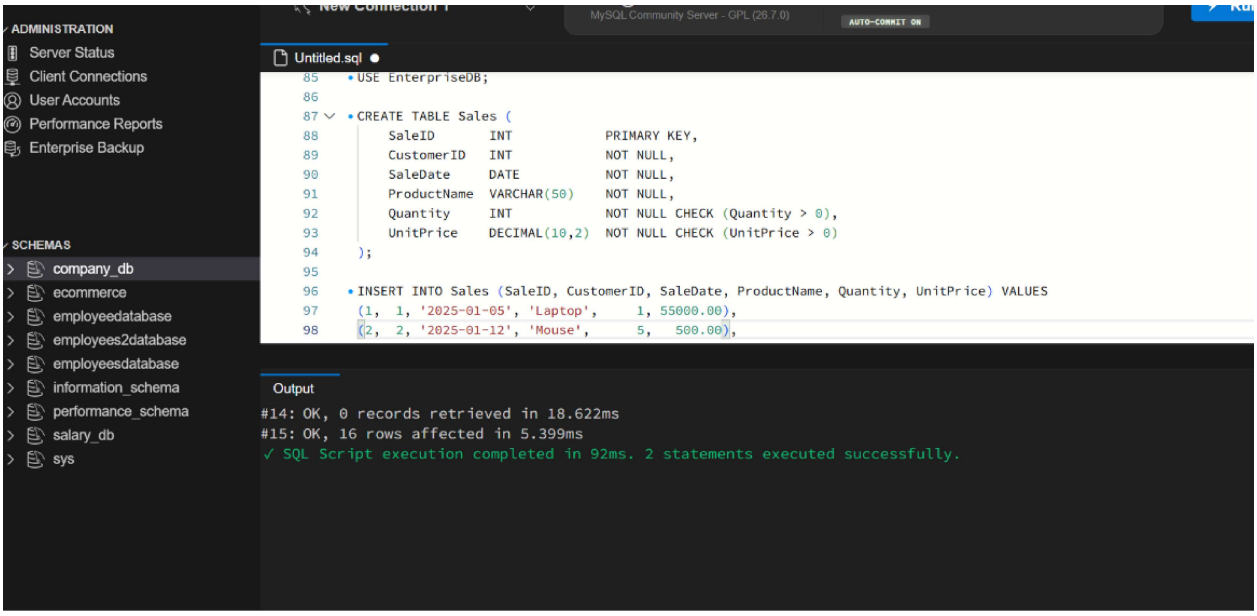


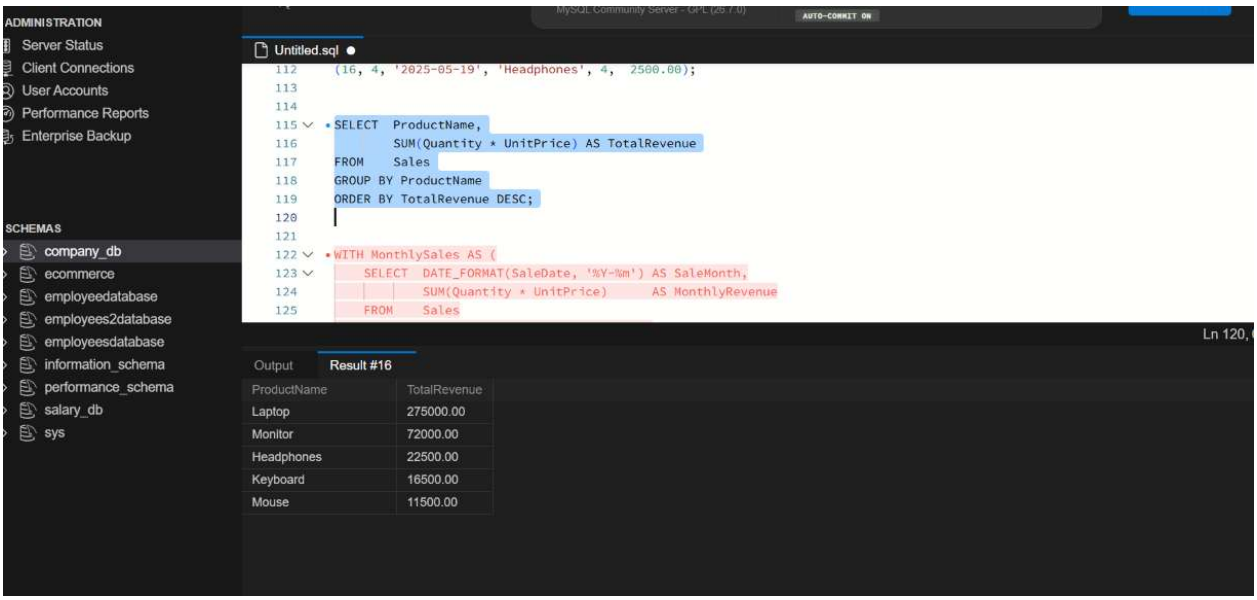

New Connection 1
root@localhost:3306
MySQL Community Server - GPL (26.7.0)
Connected
AUTO-COMMIT ON
Run
ADMINISTRATION
Server Status
Client Connections
User Accounts
Performance Reports
Enterprise Backup
SCHEMAS
company_db
ecommerce
employeedatabase
employees2database
employeesdatabase
information_schema
performance_schema
salary_db
sys
Untitled.sql
(106, 'Ravi Shankar', 1, 65000.00, '2020-08-25', 101),
(107, 'Kiran Patil', 1, 55000.00, '2022-02-14', 105),
(108, 'Divya Menon', 2, 60000.00, '2018-11-30', 102),
(109, 'Suresh Reddy', 2, 45000.00, '2021-07-19', 108),
(110, 'Pooja Sharma', 3, 90000.00, '2018-05-22', 103),
(111, 'Vikram Singh', 3, 70000.00, '2020-12-01', 103),
(112, 'Neha Joshi', 3, 50000.00, '2023-01-09', 110),
(113, 'Arjun Das', 4, 75000.00, '2019-10-17', 104),
(114, 'Lakshmi Bhat', 4, 58000.00, '2022-06-03', 104),
(115, 'Rohan Mehta', 4, 48000.00, '2023-09-28', 113);
SELECT * FROM Departments1;
SELECT * FROM Employees;
Ln 50, Col 25
Output
Result #8
EmployeeID EmployeeName DepartmentID Salary JoiningDate ManagerID
101 Anil Kumar 1 150000.00 03/10/2015 NULL
102 Sunita Rao 2 120000.00 06/15/2016 NULL
103 Rajesh Nair 3 140000.00 01/20/2014 NULL
104 Meera Iyer 4 125000.00 09/05/2017 NULL
105 Asha Gowda 1 80000.00 04/12/2019 101
106 Ravi Shankar 1 65000.00 08/25/2020 101
107 Kiran Patil 1 55000.00 02/14/2022 105
108 Divya Menon 2 60000.00 11/30/2018 102
109 Suresh Reddy 2 45000.00 07/19/2021 108

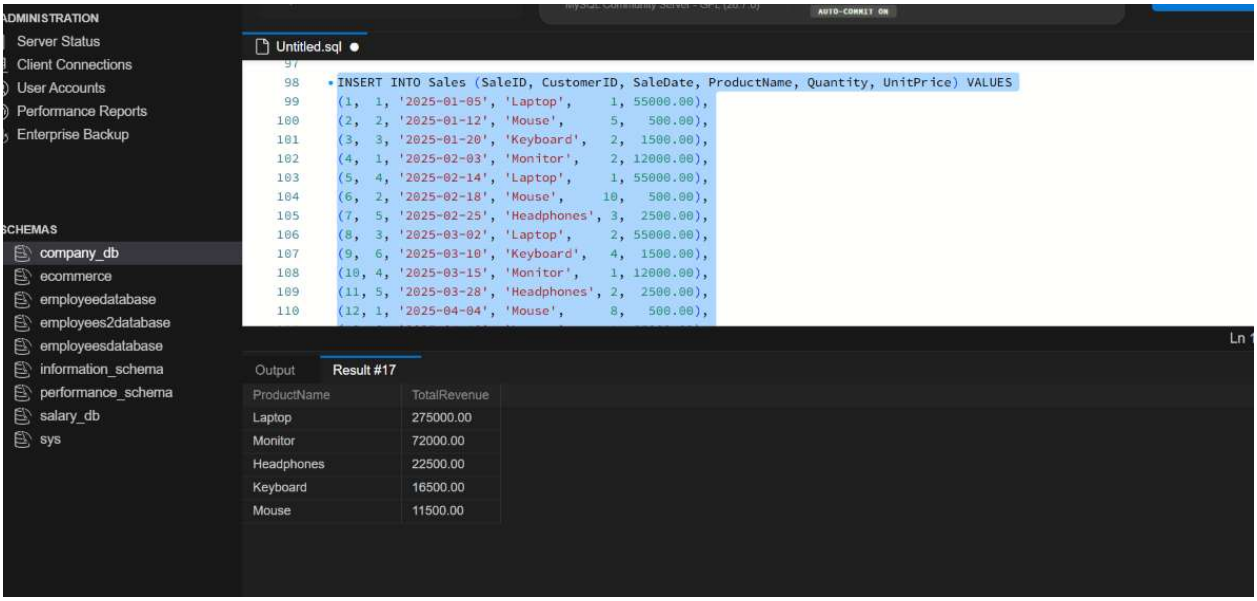

ADMINISTRATION
Server Status
Client Connections
User Accounts
Performance Reports
Enterprise Backup
SCHEMAS
company_db
ecommerce
employeedatabase
employees2database
employeesdatabase
information_schema
performance_schema
salary_db
sys
Search DB objects
Untitled.sql
(106, 'Ravi Shankar', 1, 65000.00, '2020-08-25', 101),
(107, 'Kiran Patil', 1, 55000.00, '2022-02-14', 105),
(108, 'Divya Menon', 2, 60000.00, '2018-11-30', 102),
(109, 'Suresh Reddy', 2, 45000.00, '2021-07-19', 108),
(110, 'Pooja Sharma', 3, 90000.00, '2018-05-22', 103),
(111, 'Vikram Singh', 3, 70000.00, '2020-12-01', 103),
(112, 'Neha Joshi', 3, 50000.00, '2023-01-09', 110),
(113, 'Arjun Das', 4, 75000.00, '2019-10-17', 104),
(114, 'Lakshmi Bhat', 4, 58000.00, '2022-06-03', 104),
(115, 'Rohan Mehta', 4, 48000.00, '2023-09-28', 113);
SELECT * FROM Departments1;
SELECT * FROM Employees;
Ln 50,
Output
Result #8
EmployeeID EmployeeName DepartmentID Salary JoiningDate ManagerID
106 Ravi Shankar 1 65000.00 08/25/2020 101
107 Kiran Patil 1 55000.00 02/14/2022 105
108 Divya Menon 2 60000.00 11/30/2018 102
109 Suresh Reddy 2 45000.00 07/19/2021 108
110 Pooja Sharma 3 90000.00 05/22/2018 103
111 Vikram Singh 3 70000.00 12/01/2020 103
112 Neha Joshi 3 50000.00 01/09/2023 110
113 Arjun Das 4 75000.00 10/17/2019 104
114 Lakshmi Bhat 4 58000.00 06/03/2022 104
115 Rohan Mehta 4 48000.00 09/28/2023 113

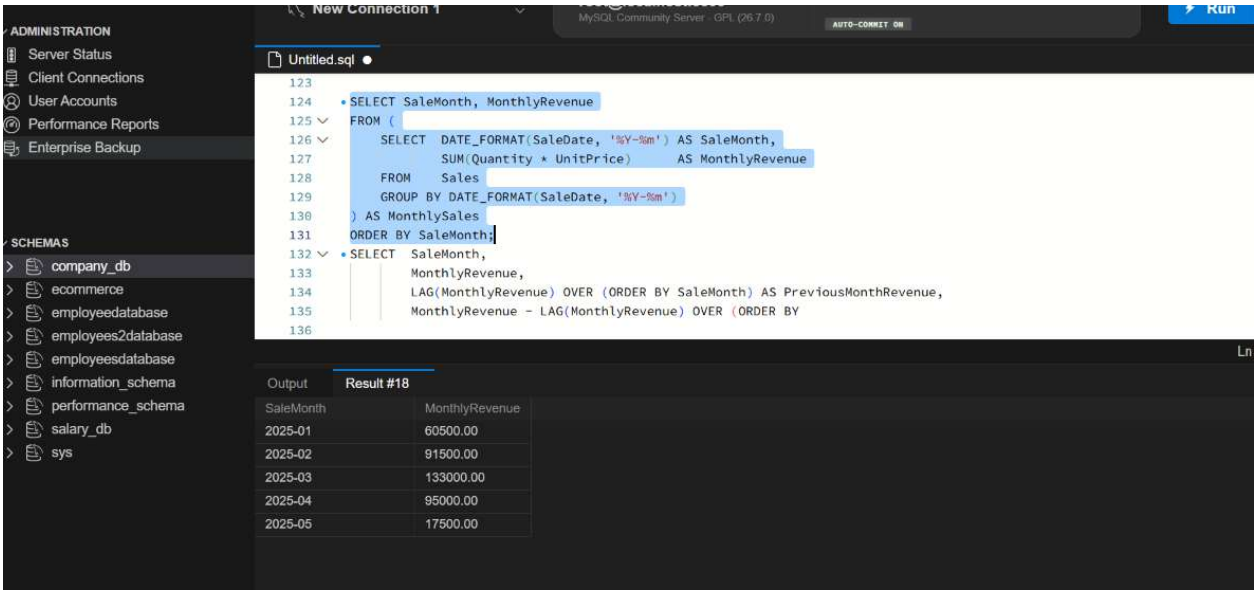

New Connection 1
MySQL Community Server - GPL (26.7.0)
AUTO-COMMIT ON
Run
ADMINISTRATION
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SCHEMAS
company_db
ecommerce
employeedatabase
employees2database
employeesdatabase
information_schema
performance_schema
salary_db
sys
Untitled.sql
59 ORDER BY d.DepartmentName, e.EmployeeName;
60
61 SELECT e.EmployeeID, e.EmployeeName, e.DepartmentID, e.Salary
62 FROM Employees1 e
63 WHERE e.Salary > (SELECT AVG(x.Salary)
64 FROM Employees1 x
65 WHERE x.DepartmentID = e.DepartmentID)
66 ORDER BY e.DepartmentID, e.Salary DESC;
67
68 SELECT MAX(Salary) AS SecondHighestSalary
69 FROM Employees1
70 WHERE Salary < (SELECT MAX(Salary) FROM Employees1);
71
72
Ln 70, Co
Output
Result #11
SecondHighestSalary
140000.00

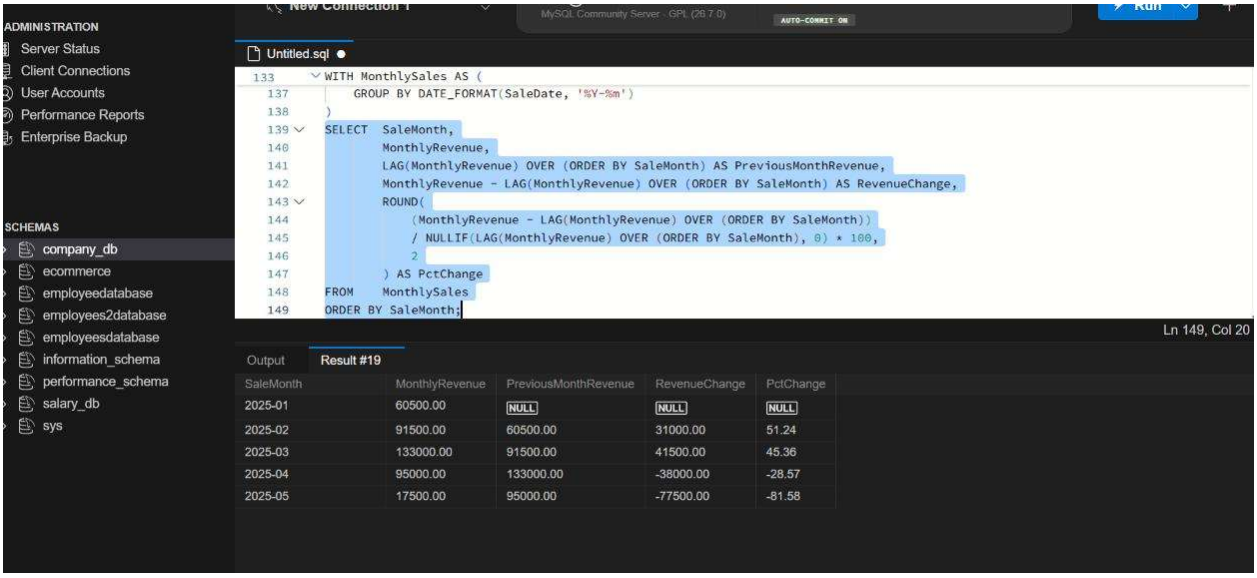

New Connection 1
MySQL Community Server - GPL (26.7.0)
AUTO-COMMIT ON
Run
ADMINISTRATION
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SCHEMAS
company_db
ecommerce
employeedatabase
employees2database
employeesdatabase
information_schema
performance_schema
salary_db
sys
Untitled.sql
71
72 SELECT DepartmentName, EmployeeName, Salary, SalaryRank
73 FROM (
74 SELECT d.DepartmentName,
75 e.EmployeeName,
76 e.Salary,
77 RANK() OVER (PARTITION BY e.DepartmentID
78 ORDER BY e.Salary DESC) AS SalaryRank
79 FROM Employees1 e
80 JOIN Departments1 d ON d.DepartmentID = e.DepartmentID
Ln 83, Col 51
Output
Result #12
DepartmentName | EmployeeName | Salary | SalaryRank
Finance | Rajesh Nair | 140000.00 | 1
Finance | Pooja Sharma | 90000.00 | 2
Finance | Vikram Singh | 70000.00 | 3
HR | Sunita Rao | 120000.00 | 1
HR | Divya Menon | 60000.00 | 2
HR | Suresh Reddy | 45000.00 | 3
IT | Anil Kumar | 150000.00 | 1
IT | Asha Gowda | 80000.00 | 2
IT | Ravi Shankar | 65000.00 | 3
Marketing | Meera Iyer | 125000.00 | 1
Marketing | Arjun Das | 75000.00 | 2
Marketing | Lakshmi Bhat | 58000.00 | 3

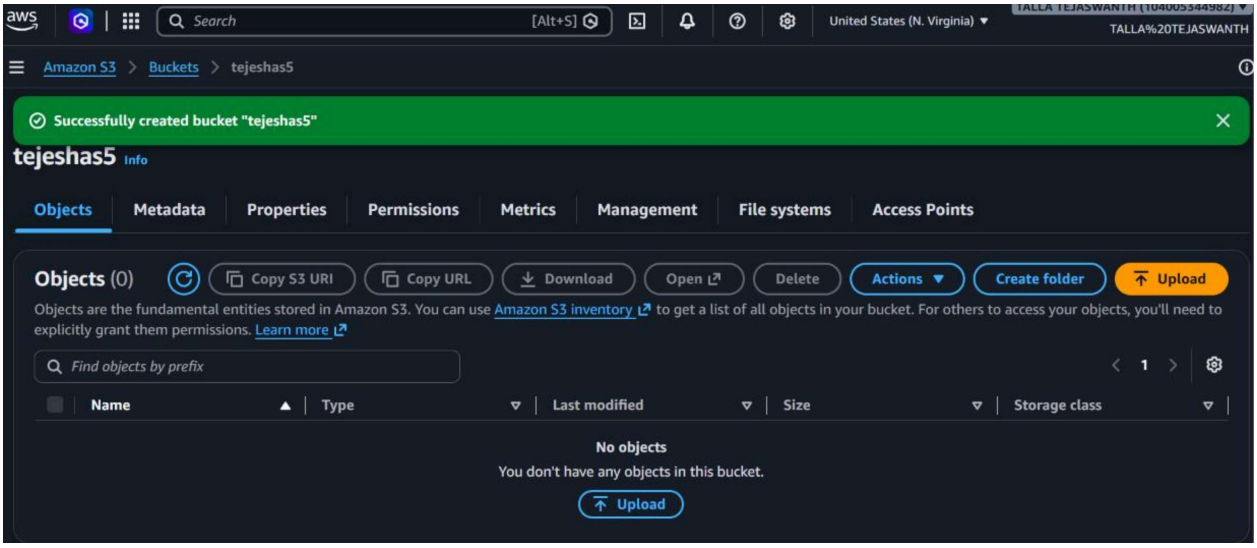

ADMINISTRATION
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ecommerce
employeedatabase
employees2database
employeesdatabase
information_schema
performance_schema
salary_db
sys
MySQL Community Server - GPL (26.7.0)
AUTO-COMMIT ON
Untitled.sql
85 USE EnterpriseDB;
86
87 CREATE TABLE Sales (
88 SaleID INT PRIMARY KEY,
89 CustomerID INT NOT NULL,
90 SaleDate DATE NOT NULL,
91 ProductName VARCHAR(50) NOT NULL,
92 Quantity INT NOT NULL CHECK (Quantity > 0),
93 UnitPrice DECIMAL(10,2) NOT NULL CHECK (UnitPrice > 0)
94);
95
96 INSERT INTO Sales (SaleID, CustomerID, SaleDate, ProductName, Quantity, UnitPrice) VALUES
97 (1, 1, '2025-01-05', 'Laptop', 1, 55000.00),
98 (2, 2, '2025-01-12', 'Mouse', 5, 500.00),
Output
#14: OK, 0 records retrieved in 18.622ms
#15: OK, 16 rows affected in 5.399ms
✓ SQL Script execution completed in 92ms. 2 statements executed successfully.

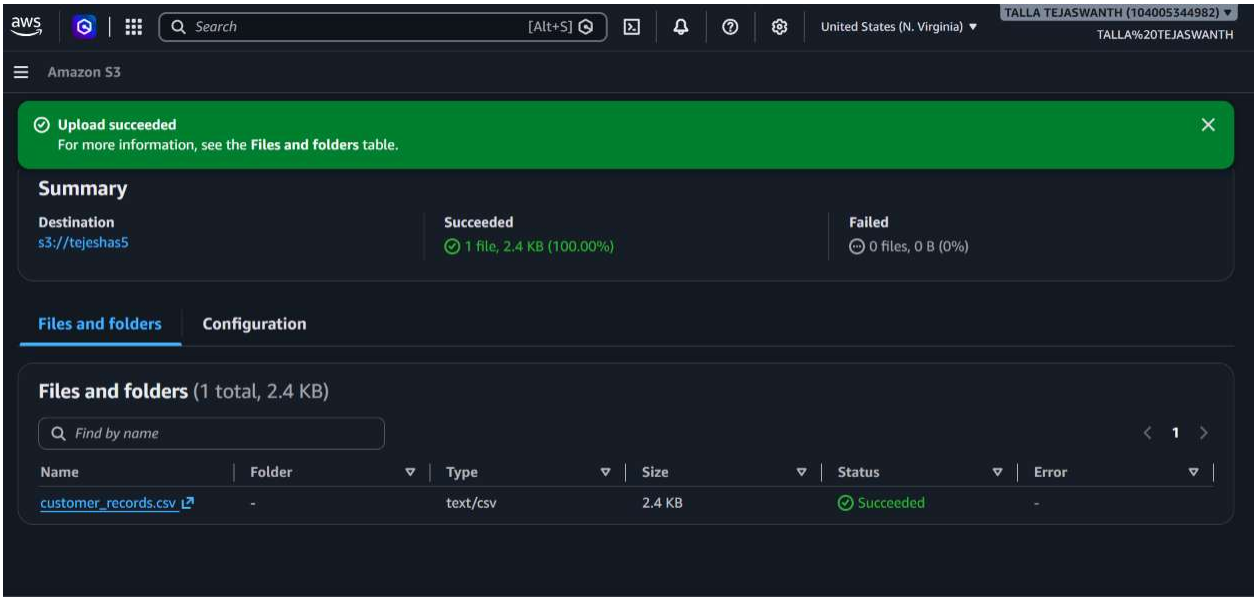

ADMINISTRATION
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company_db
ecommerce
employeedatabase
employees2database
employeesdatabase
information_schema
performance_schema
salary_db
sys
AUTO-COMMIT ON
Untitled.sql
112 (16, 4, '2025-05-19', 'Headphones', 4, 2500.00);
113
114
115 SELECT ProductName,
116 SUM(Quantity * UnitPrice) AS TotalRevenue
117 FROM Sales
118 GROUP BY ProductName
119 ORDER BY TotalRevenue DESC;
120
121
122 WITH MonthlySales AS (
123 SELECT DATE_FORMAT(SaleDate, '%Y-%m') AS SaleMonth,
124 SUM(Quantity * UnitPrice) AS MonthlyRevenue
125 FROM Sales
Ln 120,
Output
Result #16
ProductName TotalRevenue
Laptop 275000.00
Monitor 72000.00
Headphones 22500.00
Keyboard 16500.00
Mouse 11500.00

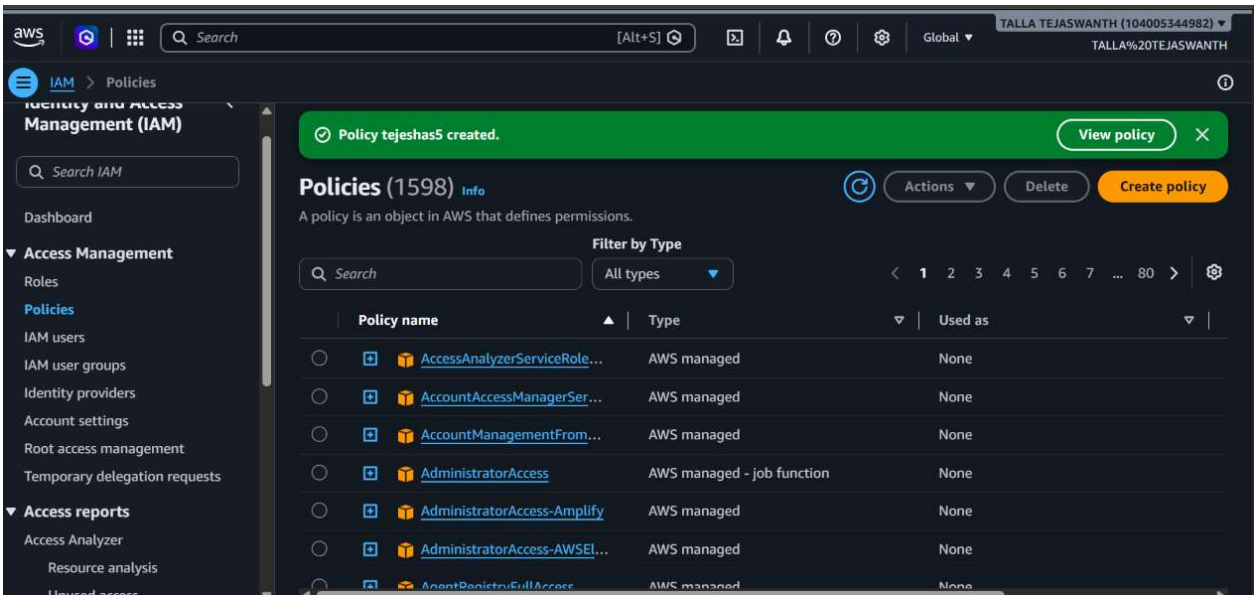

ADMINISTRATION
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SCHEMAS
company_db
ecommerce
employeedatabase
employees2database
employeesdatabase
information_schema
performance_schema
salary_db
sys
Untitled.sql
INSERT INTO Sales (SaleID, CustomerID, SaleDate, ProductName, Quantity, UnitPrice) VALUES
(1, 1, '2025-01-05', 'Laptop', 1, 55000.00),
(2, 2, '2025-01-12', 'Mouse', 5, 500.00),
(3, 3, '2025-01-20', 'Keyboard', 2, 1500.00),
(4, 1, '2025-02-03', 'Monitor', 2, 12000.00),
(5, 4, '2025-02-14', 'Laptop', 1, 55000.00),
(6, 2, '2025-02-18', 'Mouse', 10, 500.00),
(7, 5, '2025-02-25', 'Headphones', 3, 2500.00),
(8, 3, '2025-03-02', 'Laptop', 2, 55000.00),
(9, 6, '2025-03-10', 'Keyboard', 4, 1500.00),
(10, 4, '2025-03-15', 'Monitor', 1, 12000.00),
(11, 5, '2025-03-28', 'Headphones', 2, 2500.00),
(12, 1, '2025-04-04', 'Mouse', 8, 500.00),
Output
Result #17
ProductName
TotalRevenue
Laptop
275000.00
Monitor
72000.00
Headphones
22500.00
Keyboard
16500.00
Mouse
11500.00


New Connection 1
MySQL Community Server - GPL (26.7.0)
AUTO-COMMIT ON
Run
ADMINISTRATION
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SCHEMAS
company_db
ecommerce
employeedatabase
employees2database
employeesdatabase
information_schema
performance_schema
salary_db
sys
Untitled.sql
SELECT SaleMonth, MonthlyRevenue
FROM (
SELECT DATE_FORMAT(SaleDate, '%Y-%m') AS SaleMonth,
SUM(Quantity * UnitPrice) AS MonthlyRevenue
FROM Sales
GROUP BY DATE_FORMAT(SaleDate, '%Y-%m')
) AS MonthlySales
ORDER BY SaleMonth;
SELECT SaleMonth,
MonthlyRevenue,
LAG(MonthlyRevenue) OVER (ORDER BY SaleMonth) AS PreviousMonthRevenue,
MonthlyRevenue - LAG(MonthlyRevenue) OVER (ORDER BY
Output
Result #18
SaleMonth
MonthlyRevenue
2025-01
60500.00
2025-02
91500.00
2025-03
133000.00
2025-04
95000.00
2025-05
17500.00


ADMINISTRATION
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employeesdatabase
information_schema
performance_schema
salary_db
sys
Untitled.sql
WITH MonthlySales AS (
GROUP BY DATE_FORMAT(SaleDate, '%Y-%m')
)
SELECT SaleMonth,
MonthlyRevenue,
LAG(MonthlyRevenue) OVER (ORDER BY SaleMonth) AS PreviousMonthRevenue,
MonthlyRevenue - LAG(MonthlyRevenue) OVER (ORDER BY SaleMonth) AS RevenueChange,
ROUND(
(MonthlyRevenue - LAG(MonthlyRevenue) OVER (ORDER BY SaleMonth))
/ NULLIF(LAG(MonthlyRevenue) OVER (ORDER BY SaleMonth), 0) * 100,
2
) AS PctChange
FROM MonthlySales
ORDER BY SaleMonth;
Ln 149, Col 20
Output
Result #19
SaleMonth | MonthlyRevenue | PreviousMonthRevenue | RevenueChange | PctChange
2025-01 | 60500.00 | NULL | NULL | NULL
2025-02 | 91500.00 | 60500.00 | 31000.00 | 51.24
2025-03 | 133000.00 | 91500.00 | 41500.00 | 45.36
2025-04 | 95000.00 | 133000.00 | -38000.00 | -28.57
2025-05 | 17500.00 | 95000.00 | -77500.00 | -81.58


Search
[Alt+S]
United States (N. Virginia)
TALLA%20TEJASWANTH
Amazon S3 > Buckets > tejeshas5
Successfully created bucket "tejeshas5"
tejeshas5 Info
Objects
Metadata
Properties
Permissions
Metrics
Management
File systems
Access Points
Objects (0)
Copy S3 URI
Copy URL
Download
Open
Delete
Actions
Create folder
Upload
Objects are the fundamental entities stored in Amazon S3. You can use Amazon S3 inventory to get a list of all objects in your bucket. For others to access your objects, you'll need to explicitly grant them permissions. Learn more
Find objects by prefix
Name
Type
Last modified
Size
Storage class
No objects
You don't have any objects in this bucket.
Upload


Search
[Alt+S]
United States (N. Virginia)
TALLA TEJASWANTH (104005344982)
TALLA%20TEJASWANTH
Amazon S3
Upload succeeded
For more information, see the Files and folders table.
Summary
Destination
s3://tejeshas5
Succeeded
1 file, 2.4 KB (100.00%)
Failed
0 files, 0 B (0%)
Files and folders
Configuration
Files and folders (1 total, 2.4 KB)
Find by name
Name
Folder
Type
Size
Status
Error
customer_records.csv
-
text/csv
2.4 KB
Succeeded
-


Search
[Alt+S]
Global
TALLA TEJASWANTH (104005344982)
TALLA%20TEJASWANTH
IAM
Policies
Identity and Access Management (IAM)
Search IAM
Dashboard
Access Management
Roles
Policies
IAM users
IAM user groups
Identity providers
Account settings
Root access management
Temporary delegation requests
Access reports
Access Analyzer
Resource analysis
Policy tejeshas5 created.
View policy
Policies (1598) Info
A policy is an object in AWS that defines permissions.
Actions
Delete
Create policy
Filter by Type
Search
All types
1 2 3 4 5 6 7 ... 80
Policy name
Type
Used as
AccessAnalyzerServiceRole...
AWS managed
None
AccountAccessManagerSer...
AWS managed
None
AccountManagementFrom...
AWS managed
None
AdministratorAccess
AWS managed - job function
None
AdministratorAccess-Amplify
AWS managed
None
AdministratorAccess-AWSEl...
AWS managed
None

File Edit View H1

SQL TROUBLESHOOTING

A.Excecution plan shows how a data base executes an sql theory
table scans: reading the entire table
expensive joins:combining large tables
missing indexies:causing slow data retrival

B.Create a composite index on DepartmentID and Salary.

```
 CREATE INDEX idx_dept_salary
 ON Employees (DepartmentID, Salary);
```

The index improves queries filtering by DepartmentID and Salary.
It helps the database find employees in a particular department and filter them by salary.
The column order is suitable when queries filter by DepartmentID alone or by both columns.

C.

```
SELECT Email, COUNT(*) AS DuplicateCount
FROM Customers
GROUP BY Email
HAVING COUNT(*) > 1;
```

